

BFO 2020 Existence Instantiation Axioms

Particulars exist at some time [nmq-1]

$$\forall p(\text{particular}(p) \rightarrow \exists t \text{existsAt}(p,t))$$

Every universal is instantiated at least once [mbf-1]

$$\forall u(\text{universal}(u) \rightarrow \exists p,t \text{instanceOf}(p,u,t))$$

Exists at is disjunctive on first argument when it is a continuant [uns-1]

$$\forall p,q,r(\text{existsAt}(p,q) \wedge \text{continuantPartOf}(r,p,q) \rightarrow \text{existsAt}(r,q))$$

Instance of is disjunctive on third argument, a temporal region [qaf-1]

$$\forall p,q,r,s(\text{instanceOf}(p,q,r) \wedge \text{temporalPartOf}(s,r) \rightarrow \text{instanceOf}(p,q,s))$$

Relata of exists at are particulars. [oap-1]

$$\forall i,t(\text{existsAt}(i,t) \rightarrow \text{particular}(i) \wedge \text{particular}(t) \wedge \text{instanceOf}(t,\text{temporalRegion},t))$$

Relata of instance of are particular, universal, temporal region. [lqn-1]

$$\forall i,u,t(\text{instanceOf}(i,u,t) \rightarrow \text{particular}(i) \wedge \text{universal}(u) \wedge \text{instanceOf}(t,\text{temporalRegion},t))$$

There is always something that exists [nis-1]

$$\forall t(\text{instanceOf}(t,\text{temporalRegion},t) \rightarrow \exists u,i(i \neq t \wedge \text{universal}(u) \wedge \text{particular}(i) \wedge \text{instanceOf}(i,u,t)))$$

If m is a material entity, then there is some one dimensional temporal region during which m exists [zuw-1]

$$\forall m(\exists t \text{instanceOf}(m,\text{materialEntity},t) \rightarrow \exists t(\text{instanceOf}(t,\text{oneDimensionalTemporalRegion},t) \wedge \text{existsAt}(m,t)))$$

If you exist you instantiate a universal and vice versa [bee-1]

$$\forall a,t(\exists u(\text{universal}(u) \wedge \text{instanceOf}(a,u,t) \wedge \text{instanceOf}(t,\text{temporalRegion},t)) \leftrightarrow \text{particular}(a) \wedge \text{instanceOf}(t,\text{temporalRegion},t) \wedge \text{existsAt}(a,t))$$

